

View Detection - Technical Explanation

Automated Drawing Detection & Crop Generation from PDF Sheets

1. Objective

To automatically detect and extract individual drawing views — plans, elevations, sections, and details — from multi-drawing PDF sheets, without relying on any manual labeling or a single fixed detection setting that has to be tuned by hand for every sheet.

2. Problem Statement

A single PDF sheet can contain anywhere from one to a dozen or more separate drawings, laid out at different densities. Using one fixed AI confidence threshold across every sheet creates a trade-off: a strict setting misses drawings on busy sheets, while a loose setting produces false positives on simple ones. In addition, AI-generated bounding boxes are typically too tight, cutting off drawing titles, grid bubbles, or level markers — and there is normally no ground-truth data available to check detection accuracy automatically.

3. Proposed Approach

We propose a pipeline that combines an open-vocabulary AI detector (Grounding DINO) with the PDF's own vector and text geometry, so that the AI handles visual detection while the PDF's real structure handles precision and validation. The core idea is to make every key decision — the confidence threshold, the scan mode, and the crop boundaries — adapt automatically to what is actually on each sheet, rather than using one-size-fits-all settings.

4. Methodology

The approach is organized into eight sequential phases:

#	Phase	Description
1	Configuration	Centralize all tunable values (thresholds, size limits, tiling settings) in one place, with command-line overrides.
2	Model Loading	Load Grounding DINO once, cache it in memory, and run on GPU when available, CPU otherwise.
3	Adaptive Thresholding	Automatically choose a confidence cutoff and scan mode (single-pass vs. tiled) based on how many drawings are on the sheet.
4	Detection	Run Grounding DINO over the page (or its tiles), collecting a wide pool of raw candidate boxes.
5	Quality Filtering & Deduplication	Remove implausible boxes and merge duplicate detections of the same drawing.
6	Scale-Anchor Validation	Cross-check every detection against scale/title text extracted from the PDF, to measure accuracy without manual labels.
7	Geometry-Based Cropping	Build final crop boxes from the PDF's real lines and text, correcting for the

		AI's tendency to clip titles or markers.
8	Output Generation	Save overlay images, individual crops, and an aggregate results file (CSV/JSON) summarizing every page.

4.1 Adaptive Thresholding — the central mechanism

Rather than scoring detections against one fixed cutoff, the pipeline first estimates how many drawing views are on the sheet by counting scale/title notes in the PDF's text layer. Sheets with five or fewer views are scanned in a single pass with a normal threshold; sheets with six or more are split into overlapping tiles and scanned with a lower threshold, since each tile gives the AI less surrounding context to work with. When a sheet has no readable text at all (for example, a scanned image), a quick trial detection pass stands in for the view count. If the automatic choice ever yields zero results, the pipeline retries in tiled mode as a safety net.

4.2 Geometry-Based Crop Generation

Once drawings are detected, crop boundaries are not taken directly from the AI's boxes. Instead, each detection is matched to the nearest title/scale note below it, paired with the horizontal rule line printed near that title, and grown outward to capture any nearby grid bubbles or level markers. This produces crop boxes that reflect the true extent of each drawing, rather than the AI's tighter — and often incomplete — estimate.

4.3 Validation Against Scale Anchors

Every detection is checked against nearby scale/title text: exactly one match nearby is scored as a clean detection, more than one indicates two drawings may have been merged into a single box, and no match at all flags a likely false positive. This gives an automatic accuracy signal — recall, spurious, and merged rates — without requiring anyone to hand-label the sheets.

5. Key Advantages of This Approach

Advantage	Why It Matters
No fixed threshold	One sheet-wide confidence cutoff never fits every layout. Adapting it per sheet reduces both missed drawings and false positives.
No manual ground truth needed	Scale-note text already on the sheet acts as a built-in accuracy check, so quality can be measured automatically on every run.
Geometry-corrected crops	Using the PDF's real lines and text avoids the clipped titles and grid markers typical of raw AI bounding boxes.
Built-in safety net	If the automatic mode choice turns out wrong for an unusual sheet, the pipeline retries automatically rather than silently failing.
Fully configurable	Every threshold and tolerance is exposed as a command-line option, so the approach can be tuned without changing code.

6. Configurability

Every threshold, tolerance, and pipeline behavior is exposed as a command-line option, so the approach can be adjusted for different sheet styles or accuracy requirements without any code changes:

```
python dino_lab_adaptive.py <pdf> [options]
```

- Scan mode: automatic, single-pass, tiled, or both
- Thresholds: single-pass, tiled, and fallback-retry values
- Thresholding strategy: sheet-profile (default) or score-gap
- Duplicate-merge sensitivity, crop generation on/off, and per-crop image export
- Rendering resolution, tile size and overlap, and output location

7. Deliverables

- An annotated overlay image per page, showing detected drawings, scale anchors, and crop boxes
- Individual cropped image files for each detected drawing (optional)
- A per-page JSON record of detections, crops, and scores
- An aggregate CSV/JSON summary across all processed pages, including threshold used, detection counts, and accuracy metrics

8. Conclusion

This approach avoids the need for a general-purpose visual-language model or any manually labeled training data. All contextual understanding comes from combining Grounding DINO's visual detection with text and geometry already present in the PDF. By letting the confidence threshold, scan mode, and crop boundaries adapt to each sheet automatically — and by building in a retry safety net — the pipeline is designed to generalize across varied drawing sheets with minimal manual tuning.